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Medical Financial Fragility

Insurance Design and Access Barriers in MEPS

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Abstract

This paper measures medical financial fragility in the 2022 Medical Expenditure Panel Survey (MEPS). The motivating idea is simple: realized medical spending alone is not a sufficient measure of household medical vulnerability because low spending can reflect low need, good coverage, or foregone care. The paper therefore studies out-of-pocket medical burden, insurance status, health need, and delayed or unaffordable care jointly. Using the MEPS Household Component Full-Year Consolidated file, the paper shows that lower-income people spend fewer dollars out of pocket than high-income people but face much higher burden relative to family income and higher access-barrier rates. Uninsured people have lower observed out-of-pocket spending than privately insured people but much higher delayed/unaffordable-care rates, reinforcing the point that low spending can signal unmet need. The revised draft now benchmarks the proposed joint fragility profile against simpler screens such as OOP ≥10 percent of income, access-barrier-only measures, and uninsured status. The comparison shows that a spending-only threshold misses a large population with access barriers and that zero observed out-of-pocket spending can coexist with delayed or unaffordable care. The analysis is descriptive and should be read as evidence on co-occurrence and risk profiles, not as causal evidence that out-of-pocket burden causes access barriers. A preliminary Person-Round-Plan construction explores private premium exposure, but the paper treats that component as exploratory pending additional premium-construction validation.

Version status

This is an advanced public working paper / MEPS measurement paper. It uses 2022 MEPS data and now checks the main findings several ways: sample restrictions, selected survey-design confidence intervals, family construction, PRPL premium linkage, denominator sensitivity, access-barrier components, model robustness, 2019–2023 robustness benchmarks, premium annualization, questionnaire continuity, and comparisons with simpler fragility and underinsurance-style screens. The family-level estimates and PRPL premium linkage remain exploratory pending additional validation. The main descriptive contribution does not depend on the exploratory premium or family tables.

1. Introduction

Medical costs are usually measured either as prices, spending, insurance coverage, or affordability problems. Each measure captures part of the household problem, but none is sufficient alone. A household with high realized out-of-pocket medical spending may be financially strained because it used needed care. A household with low spending may be protected, healthy, or unable to afford care. A household with insurance may still face deductibles, premiums, formularies, or network constraints. A household without insurance may avoid care and therefore look inexpensive in expenditure data.

This paper develops a descriptive measurement frame for this problem. I call the broader concept a medical fragility burden: the joint exposure created when medical cost-sharing, insurance design, health need, and household resources interact with access to care. The term is deliberately broader than out-of-pocket spending. It is intended to separate three objects that are often blended: realized medical cost-sharing, insurance-design exposure, and delayed or unaffordable care.

The paper uses the 2022 Medical Expenditure Panel Survey (MEPS) Household Component. The main empirical exercise links person-level out-of-pocket medical spending, family income, poverty category, annual insurance coverage, self-reported health, and access-barrier outcomes. It also builds an exploratory private-insurance premium measure from the MEPS Person-Round-Plan file. The premium component is promising but not yet validated enough to carry the paper.

The current contribution is measurement, not identification. The paper does not estimate the causal effect of medical burden on access barriers. Out-of-pocket spending, health, income, insurance, and access barriers are jointly determined. Instead, the paper documents the joint distribution of medical financial fragility and asks whether a joint burden-design-need-access profile adds information beyond simpler screens such as high out-of-pocket burden, uninsured status, or access barriers alone.

The core descriptive facts are useful. People in poor or negative-income families have lower dollar out-of-pocket medical spending than high-income people, but they face much higher burden relative to income and more access barriers. People in poor health face both higher spending and more barriers. Uninsured people do not have the highest realized out-of-pocket spending, but they have the highest access-barrier rates. These patterns are not surprising in isolation. Their value is that they show why spending, coverage, and access must be interpreted together.

2. Relationship to existing research

This paper sits between household finance and health economics. Household-finance research on financial fragility emphasizes liquid resources and the ability to absorb shocks. Medical-shock research shows that health events can affect out-of-pocket spending, unpaid bills, bankruptcy risk, employment, and earnings. Health-services research on underinsurance, high-deductible plans, medical debt, cost-related nonadherence, and delayed care documents how insurance design and household resources shape access.

The paper does not claim that medical affordability is a new topic. Its measurement claim is narrower: a useful household-fragility measure should jointly observe burden, design, need, and access. Standard out-of-pocket burden thresholds are informative but can miss households that avoid care. Insurance status is informative but cannot distinguish generous coverage from high-deductible exposure. Access barriers are informative but do not by themselves reveal financial burden. The contribution is to organize these dimensions as one risk profile and to test whether MEPS can support that measurement structure.

This positioning also clarifies what the paper is not. It is not a causal Medicaid-expansion paper, a high-deductible-plan event study, or a structural model of healthcare demand. Those are plausible future directions. The present draft is a measurement paper that asks whether public MEPS microdata can identify medical financial fragility more credibly than aggregate price or spending proxies.

3. Data, variables, and estimands

The main data source is the 2022 MEPS Household Component Full-Year Consolidated Public Use File, HC-243. The proof-of-concept build loads 22,431 person records. The main person weight is `PERWT22F`; survey design variables `VARSTR` and `VARPSU` are retained and used for selected validation confidence intervals. The companion source is HC-242, the Person-Round-Plan file, used only for exploratory private premium construction.

The primary person-weighted estimand is: what share of people live in families with a given medical out-of-pocket burden and access-barrier profile? This is distinct from a family-weighted estimand: what share of families face a given burden? Both are valid. The current draft uses person-level estimates as the main analysis and includes a preliminary family-level table as a diagnostic. A journal version should rebuild the family-level analysis according to MEPS family-estimation guidance and pre-specify whether person-weighted or family-weighted estimates are primary.

The core burden measure is annual out-of-pocket medical spending divided by family income for positive-family-income records. Because ratios can be unstable for very low income, the paper also reports threshold measures: out-of-pocket spending at least 5, 10, or 20 percent of family income, and a version of the burden mean capped at 100 percent. These are robustness measures, not replacements for full denominator work.

Table S1. MEPS 2022 sample construction and analytic restrictions. The table documents why sample sizes differ across descriptive, burden-ratio, regression, PRPL-linked, and family-level checks. Burden-ratio rows require positive family income; PRPL and family-level rows are exploratory extensions.

Stage	Unweighted N	Weighted pop./families	Purpose / restriction
Full HC-243 person records loaded	22,431	333.1M	All person records in the 2022 Full-Year Consolidated PUF extract.
Positive-family-income burden sample	21,603	324.4M	Required for OOP/income ratios and burden quartiles; excludes zero/negative/missing family income.
Access-barrier composite observed	22,050	328.9M	Any delayed/unaffordable medical, dental, or prescription measure observed.
Latest-round health status observed	22,073	328.4M	Required for health-status descriptive tables.
Regression analytic sample	17,043	256.6M	Complete cases for burden quartile, access outcome, demographics, insurance, health, employment, family size, race/ethnicity, region, and person weight.
PRPL-linked privately insured sample	9,555	167.8M	Privately insured persons assigned a positive exploratory family premium from HC-242.
Collapsed draft family records	11,244	—	Approximate `DUID:FAMIDYR` family key with `FAMID22` fallback; used only as a secondary family-level check.
Family records with positive family weight	10,033	142.4M	Family-weighted sample for the secondary check; not yet validated against full MEPS family-estimation guidance.

Table A1. Core MEPS variable construction. Variables are from HC-243 except PRPL premium fields from HC-242. Negative MEPS inapplicable/missing codes are treated as missing in analytic variables.

Construct	MEPS source variable(s)	Draft construction	Main caveat
Medical OOP spending	`TOTSLF22`	Annual self/family payment for medical care	Realized spending can be low because care was foregone
Family income	`FAMINC22`	Family income denominator	Zero/negative/very low income creates unstable ratios
OOP burden	`TOTSLF22 / FAMINC22`	Person-level burden for positive-family-income records	Person-weighted, family income repeated across family members
Threshold burden	`TOTSLF22 / FAMINC22`	Indicators for OOP ≥5%, ≥10%, ≥20% of income	Less tail-sensitive than raw burden means
Annual insurance status	`INSCOV22`	Any private, public only, uninsured	Broad annual categories; does not capture plan generosity
Physical health	`RTHLTH53`	Excellent, very good, good, fair, poor; fair/poor indicator	Latest-round self-report; not a causal health control
Access-barrier composite	`DLAYCA42`, `AFRDCA42`, `DLAYDN42`, `AFRDDN42`, `DLAYPM42`, `AFRDPM42`	Any delayed/unaffordable medical, dental, or prescription care	Round timing and eligibility require full MEPS documentation in journal version
Medical barrier	`DLAYCA42`, `AFRDCA42`	Any delayed/unaffordable medical care	Component of composite

Construct	MEPS source variable(s)	Draft construction	Main caveat
Dental barrier	`DLAYDN42`, `AFRDDN42`	Any delayed/unaffordable dental care	Dental coverage differs from medical coverage
Prescription barrier	`DLAYPM42`, `AFRDPM42`	Any delayed/unaffordable prescription medicines	May be more tightly linked to plan formularies/cost sharing
Private premium exposure	HC-242 `OOPX12X`, `PHLDRIDX`, `ESTBIDX`, `InsurPrivIDEX`, `RN`	De-duplicate policyholder-establishment-plan-round-premium records and assign to policyholder family	Exploratory until PRPL construction is fully audited
Family identifier	`DUID:FAMIDYR` fallback `FAMID22`	Approximate MEPS family key for proof-of-concept family tables	Must be rebuilt/validated against MEPS family-estimation guidance
Weights/design	`PERWT22F`, `FAMWT22F`, `VARSTR`, `VARPSU`	Person/family weights retained in extracts; approximate design-based CIs added for selected proportions	Full MEPS variance validation remains needed before journal submission

4. Descriptive evidence

4.1 Burden by poverty and income category

Table 1 shows the basic distributional pattern. Dollar spending rises with income, but burden falls. The main tables remain weighted descriptive estimates; a new validation table below adds approximate MEPS design-based confidence intervals for selected proportions, so subgroup differences should still be read as descriptive patterns rather than causal effects. People in poor or negative-income families average $375 in out-of-pocket medical spending, compared with $1,228 among high-income people. Relative to mean family income, however, those amounts are 3.40 percent and 0.68 percent, respectively. The 90th-percentile burden is 11.26 percent in the poor/negative-income group and 2.03 percent in the high-income group.

Table 1. MEPS 2022 medical out-of-pocket burden by poverty/income category. Person-level weighted estimates from HC-243. OOP is `TOTSLF22`; income is `FAMINC22`; access barrier equals any delayed/unaffordable medical, dental, or prescription care indicator observed in round 4/2.

Group	N	Weighted pop.	Mean family income	Mean OOP medical	Mean OOP / mean income %	Median OOP/income %	P90 OOP/income %	OOP ≥10% income %	Any access barrier %
Poor/negative income	3,626	38.3M	$11,029	$375	3.40	0.03	11.26	10.9	20.4
Near poor	1,017	12.5M	$26,978	$681	2.52	0.06	6.80	7.3	20.5
Low income	3,000	41.2M	$40,213	$697	1.73	0.10	5.84	5.6	18.7
Middle income	6,054	98.7M	$71,885	$814	1.13	0.18	3.67	2.4	16.0
High income	8,050	142.3M	$179,328	$1,228	0.68	0.20	2.03	0.9	10.1

The access-barrier gradient points in the same direction. About 20 percent of people in poor/negative-income and near-poor families report delayed or unaffordable medical, dental, or prescription care, compared with 10.1 percent among high-income people. The interpretation is descriptive: burden and barriers cluster more heavily in lower-resource groups.

4.2 Insurance status and the ambiguity of low spending

Table 2 is central to the paper. Privately insured people have higher mean out-of-pocket spending than uninsured people, but uninsured people have much higher access barriers.

Table 2. MEPS 2022 medical out-of-pocket burden by annual insurance coverage. Person-level weighted estimates from HC-243. Insurance labels use standard MEPS annual coverage categories: any private, public only, uninsured.

Group	N	Weighted pop.	Mean family income	Mean OOP medical	Mean OOP / mean income %	Median OOP/income %	P90 OOP/income %	OOP ≥10% income %	Any access barrier %
Any private	12,723	217.3M	$130,001	$1,077	0.83	0.23	3.07	2.4	12.6
Public only	7,554	94.3M	$56,133	$650	1.16	0.05	4.39	4.7	15.2
Uninsured	1,470	21.5M	$70,240	$532	0.76	0.00	2.18	3.5	30.0

This pattern is the cleanest illustration of the measurement problem. Low observed spending among the uninsured cannot be interpreted as low vulnerability. It may reflect foregone care.

4.3 Health status and need

Table 3 shows a steep health gradient. People reporting poor health have mean out-of-pocket spending of $2,052, a 90th-percentile burden of 13.07 percent of family income, and access barriers of 35.8 percent.

Table 3. MEPS 2022 medical out-of-pocket burden by self-reported health status. Person-level weighted estimates from HC-243 using latest-round health status (`RTHLTH53`).

Group	N	Weighted pop.	Mean family income	Mean OOP medical	Mean OOP / mean income %	Median OOP/income %	P90 OOP/income %	OOP ≥10% income %	Any access barrier %
Excellent	5,434	93.4M	$127,459	$600	0.47	0.03	1.51	1.2	7.1
Very good	7,024	111.7M	$113,011	$899	0.80	0.21	2.99	2.6	12.1
Good	6,294	90.4M	$88,619	$1,060	1.20	0.29	4.07	3.8	18.9
Fair	2,152	26.4M	$69,837	$1,321	1.89	0.59	6.49	6.6	29.6
Poor	551	6.5M	$53,552	$2,052	3.83	1.04	13.07	13.4	35.8

This confirms that medical financial fragility is not only an income gradient. Health need and affordability are intertwined.

4.4 Access barriers by out-of-pocket burden quartile

Table 4 groups people by out-of-pocket burden quartile. Access barriers rise from 8.9 percent in the lowest burden quartile to 24.0 percent in the highest. Fair or poor health also rises across quartiles.

Table 4. MEPS 2022 access barriers by out-of-pocket medical burden quartile. Positive-family-income persons only. Descriptive, not causal; higher burden can reflect worse health as well as prices or coverage.

OOP burden quartile	N	Weighted pop.	Mean OOP medical	Median OOP/income %	Mean OOP/income %	Any access barrier %	Fair/poor health %
Q1	5,665	91.9M	$0	0.00	0.00	8.9	5.5
Q2	4,057	70.4M	$92	0.05	0.06	9.6	7.2
Q3	5,145	81.0M	$548	0.43	0.47	14.9	9.8
Q4	6,070	81.1M	$3,112	2.55	31.89	24.0	16.9

The high mean in the top quartile reflects strong skewness. This is substantively important, but it also motivates more robust threshold reporting.

4.5 Access-barrier components and zero spending

The access-barrier composite combines medical, dental, and prescription affordability/delay measures. Table 5 decomposes the composite by insurance status and reports zero-spending shares.

Table 5. Access-barrier components by annual insurance coverage. Person-level weighted estimates. The composite is decomposed into medical, dental, and prescription barriers; the final columns show why zero observed OOP spending cannot automatically be read as low vulnerability.

Insurance group	N	Weighted pop.	Any barrier %	Medical barrier %	Dental barrier %	Prescription barrier %	Zero OOP %	Any barrier among zero-OOP %
Any private	12,628	215.6M	12.6	5.6	9.2	2.5	19.8	6.9
Public only	7,422	92.3M	15.2	4.4	12.7	3.7	43.3	7.1
Uninsured	1,447	21.0M	30.0	18.5	23.9	6.4	57.6	22.5

The uninsured have the highest barrier rates in every component: 18.5 percent for medical barriers, 23.9 percent for dental barriers, and 6.4 percent for prescription barriers. They also have the highest zero-OOP share, 57.6 percent. Among uninsured people with zero observed out-of-pocket spending, 22.5 percent report an access barrier. This is strong descriptive evidence that zero spending and low spending require careful interpretation. The access-barrier outcomes are useful for detecting unmet need, but the exact MEPS timing and eligibility rules still need a documentation audit before these estimates should be treated as final.

4.6 Burden-threshold robustness

Table 6 reports threshold measures by poverty category. These measures are less sensitive to extreme low-income ratios than raw burden means.

Table 6. Burden-threshold robustness by poverty/income category. Person-level weighted estimates using positive-family-income records for burden ratios. Threshold measures are less sensitive to extreme low-income ratios than means.

Group	N	Weighted pop.	OOP ≥5% income %	OOP ≥10% income %	OOP ≥20% income %	Mean OOP/income capped at 100%	Zero OOP %	Any access barrier %
Poor/negative income	2,816	29.7M	16.2	10.9	5.4	4.35	47.9	20.4
Near poor	1,017	12.5M	13.1	7.3	2.7	2.74	44.7	20.5
Low income	3,000	41.2M	11.2	5.6	2.0	2.19	38.7	18.7
Middle income	6,054	98.7M	7.0	2.4	1.0	1.44	31.0	16.0
High income	8,050	142.3M	2.9	0.9	0.3	0.86	18.1	10.1

The threshold pattern preserves the main result. In the poor/negative-income group, 16.2 percent of people with positive family income have out-of-pocket medical spending of at least 5 percent of income, 10.9 percent have at least 10 percent, and 5.4 percent have at least 20 percent. In the high-income group, the corresponding rates are 2.9, 0.9, and 0.3 percent. The change in N between Table 1 and Table 6 is intentional: ratio and threshold burden measures exclude zero, negative, and missing family-income denominators. That exclusion is most consequential in the poor/negative-income category and is one reason denominator sensitivity remains a central next-step validation issue.

4.7 Exploratory private-premium burden

Table 7 adds a preliminary private-insurance premium construction from the Person-Round-Plan file. This is an exploratory appendix-style result, not part of the main burden estimate. The PRPL file is not a simple person file; premium records can repeat across dependents and plan records. The current build de-duplicates by policyholder, establishment, plan, round, and premium amount, then assigns premium exposure to the policyholder family. A validation audit below reports record-count and linkage sensitivity across plausible de-duplication keys.

Table 7. Exploratory PRPL premium-linked burden among privately insured persons — not part of the main estimate. Premiums are conservatively de-duplicated by policyholder-establishment-plan-round before family assignment. The medical OOP dollar column remains medical OOP only; burden percentile columns use combined medical OOP plus assigned PRPL premium over family income. This table is proof-of-concept evidence and should not be treated as a final premium estimate until PRPL construction is fully audited. Extreme low-income ratios are especially sensitive to denominator instability.

Group	N	Weighted pop.	Mean family income	Mean OOP medical only	Medical OOP / mean income %	Median combined OOP+premium/income %	P90 combined OOP+premium/income %	Medical OOP ≥10% income %	Any access barrier %
Poor/negative income	331	4.0M	$15,122	$632	4.18	14.22	130.70	21.7	22.0
Near poor	161	2.3M	$28,879	$849	2.94	6.03	30.89	7.9	22.9
Low income	822	12.1M	$42,414	$902	2.13	8.89	26.56	5.9	17.4
Middle income	3,315	56.4M	$76,240	$856	1.12	5.38	14.46	2.0	14.1
High income	6,205	113.0M	$185,759	$1,256	0.68	2.91	8.21	0.8	9.0

These estimates should not yet be treated as final or as a headline finding. They suggest that premium-inclusive burden may be important for lower-income privately insured households, but they still require external validation against MEPS PRPL documentation and careful denominator treatment before publication as a central result. Extreme low-income ratios are especially sensitive to denominator instability.

4.8 Preliminary family-level estimates

Table 8 repeats the burden gradient using approximate family-level construction. It is diagnostic only. I do not use Table 8 as a final family-level result; it is included to distinguish person-weighted from family-weighted estimands and to motivate a validated family-level rebuild.

Table 8. Family-level medical out-of-pocket burden by poverty/income category. Families are approximated with `DUID:FAMIDYR`; family weights are used for descriptive proof-of-concept estimates. A journal version should follow MEPS family-estimation guidance exactly.

Group	N families	Weighted families	Mean family income	Mean family OOP	Mean OOP / mean income %	Median OOP/income %	P90 OOP/income %	Any member access barrier %
Poor/negative income	1,756	18.6M	$9,323	$795	8.53	1.21	24.37	34.8
Near poor	508	5.5M	$21,937	$1,278	5.83	0.92	12.11	33.5
Low income	1,422	18.1M	$32,657	$1,502	4.60	1.16	11.90	30.9
Middle income	2,718	41.7M	$60,606	$1,876	3.10	1.08	7.53	29.0
High income	3,623	58.5M	$160,733	$2,872	1.79	0.88	4.39	18.7

The family table points in the same direction as the person-level estimates: lower-income families face higher burden shares and more access barriers. But the current family key should still be validated against MEPS family-estimation guidance before publication.

4.9 Validation checks: design uncertainty, family construction, and PRPL linkage

The May 2026 validation pass adds three checks on uncertainty, family construction, and premium linkage. They do not convert the paper into a final MEPS analysis, but they materially improve the evidentiary footing of the working paper.

Table V1. PRPL premium-linkage sensitivity audit. Premium records are filtered to hospital/physician coverage with nonmissing annual out-of-pocket premiums. The main draft strategy de-duplicates by policyholder, source, plan, round, and premium. Alternatives show sensitivity in record counts and linked private-insurance population. This validates the warning that PRPL premium estimates remain exploratory.

Strategy	Unique records	Attached records	Families	Linked private persons	Weighted linked pop.	Mean family premium
main_policyholder_source_plan_round_premium	6,998	6,582	5,621	10,917	188.8M	$4,538
policyholder_source_plan_round	6,998	6,582	5,621	10,917	188.8M	$4,538
policyholder_source_plan_premium	6,998	6,582	5,621	10,917	188.8M	$4,538
policyholder_plan_premium	6,998	6,582	5,621	10,917	188.8M	$4,538

The candidate de-duplication keys produce the same counts in this 2022 build, which is reassuring for the proof of concept. The audit does not settle PRPL annualization or official-family assignment questions, so the premium-inclusive result remains exploratory.

Table V2. MEPS design-based confidence intervals for key descriptive proportions. Estimates use person weights plus `VARSTR`/`VARPSU` in an approximate Taylor-style stratified PSU linearization for weighted proportions. Intervals are for validation and scale, not causal inference.

Panel	Group	N	Weighted pop.	Estimate % (95% CI)
Insurance	Any private	12,628	215.6M	12.6 (11.7, 13.5)
Insurance	Public only	7,422	92.3M	15.2 (13.9, 16.5)
Insurance	Uninsured	1,447	21.0M	30.0 (26.0, 34.0)
Poverty/income	Poor/negative income	2,816	29.7M	10.9 (9.2, 12.5)
Poverty/income	Near poor	1,017	12.5M	7.3 (5.0, 9.5)
Poverty/income	Low income	3,000	41.2M	5.6 (4.6, 6.7)
Poverty/income	Middle income	6,054	98.7M	2.4 (1.9, 2.8)
Poverty/income	High income	8,050	142.3M	0.9 (0.7, 1.2)
Health	Excellent	5,395	92.6M	7.1 (6.1, 8.2)
Health	Very good	7,011	111.5M	12.1 (11.1, 13.1)
Health	Good	6,277	90.1M	18.9 (17.5, 20.3)
Health	Fair	2,151	26.4M	29.6 (27.1, 32.1)
Health	Poor	549	6.5M	35.8 (30.3, 41.2)

The uncertainty table supports the main descriptive gradients. The uninsured access-barrier estimate is much higher than the private-coverage estimate; high-burden shares decline monotonically across poverty/income categories; and access barriers rise sharply as self-reported health worsens.

Table V3. Family-construction check. Families use the draft key `DUID:FAMIDYR` with `FAMID22` fallback and family weights. Missing family weights: 1,211. Person records with family-size mismatch against collapsed family count: 1,752. Empty family-year identifiers: 0. This supports descriptive checks but does not replace a full MEPS family-estimation protocol.

Poverty/income group	N families	Weighted families	N persons	Weighted persons
Poor/negative income	2,100	18.7M	3,725	38.3M
Near poor	568	5.5M	1,050	12.5M
Low income	1,579	18.1M	3,105	41.2M

Poverty/income group	N families	Weighted families	N persons	Weighted persons
Middle income	3,012	41.7M	6,269	98.7M
High income	3,985	58.5M	8,282	142.3M

The family audit shows why the person-weighted tables remain the main evidence in this draft. The family-level gradient is directionally useful, but the mismatch count and missing family weights require a careful family-estimation rebuild before family-level estimates become headline results.

4.10 Second-pass validation: denominator sensitivity, component uncertainty, and model robustness table

The May 17 review pass adds three further checks. These are designed to pressure-test the measurement claim rather than to create a causal design.

Table V4. Denominator sensitivity for MEPS out-of-pocket burden ratios. Rows compare the main positive-income denominator with increasingly conservative income floors. The exercise shows which burden gradients are stable and where very-low-income denominators drive tail ratios. Access barriers are reported alongside burden measures because unmet need remains visible even when ratio denominators are tightened.

Income denominator	Poverty/income group	N	Weighted pop.	Median OOP/income %	P90 OOP/income %	OOP ≥10% income %	Mean burden capped at 100%	Any access barrier %
Positive income	Poor/negative income	2,816	29.7M	0.03	11.26	10.9	4.35	20.9
Positive income	Near poor	1,017	12.5M	0.06	6.80	7.3	2.74	20.5
Positive income	Low income	3,000	41.2M	0.10	5.84	5.6	2.19	18.7
Positive income	Middle income	6,054	98.7M	0.18	3.67	2.4	1.44	16.0
Positive income	High income	8,050	142.3M	0.20	2.03	0.9	0.86	10.1
Income ≥ $1,000	Poor/negative income	2,745	29.0M	0.02	10.12	10.0	3.56	21.1
Income ≥ $1,000	Near poor	1,017	12.5M	0.06	6.80	7.3	2.74	20.5
Income ≥ $1,000	Low income	3,000	41.2M	0.10	5.84	5.6	2.19	18.7
Income ≥ $1,000	Middle income	6,054	98.7M	0.18	3.67	2.4	1.44	16.0
Income ≥ $1,000	High income	8,050	142.3M	0.20	2.03	0.9	0.86	10.1
Income ≥ $5,000	Poor/negative income	2,423	25.6M	0.02	6.38	7.4	2.54	20.8
Income ≥ $5,000	Near poor	1,017	12.5M	0.06	6.80	7.3	2.74	20.5
Income ≥ $5,000	Low income	3,000	41.2M	0.10	5.84	5.6	2.19	18.7
Income ≥ $5,000	Middle income	6,054	98.7M	0.18	3.67	2.4	1.44	16.0
Income ≥ $5,000	High income	8,050	142.3M	0.20	2.03	0.9	0.86	10.1
Income ≥ $10,000	Poor/negative income	1,863	19.8M	0.01	5.87	6.2	2.14	20.2
Income ≥ $10,000	Near poor	1,017	12.5M	0.06	6.80	7.3	2.74	20.5
Income ≥ $10,000	Low income	3,000	41.2M	0.10	5.84	5.6	2.19	18.7
Income ≥ $10,000	Middle income	6,054	98.7M	0.18	3.67	2.4	1.44	16.0
Income ≥ $10,000	High income	8,050	142.3M	0.20	2.03	0.9	0.86	10.1

Table V5. Approximate design-based confidence intervals for access-barrier components by insurance status. Estimates use person weights plus `VARSTR`/`VARPSU` with the same transparent Taylor-style PSU linearization as the earlier validation table. The zero-OOP rows test the central measurement warning that low realized spending can coexist with delayed or unaffordable care.

Insurance group	Measure	N	Weighted pop.	Estimate % (95% CI)
Any private	Any barrier	12,628	215.6M	12.6 (11.7, 13.5)
Any private	Medical barrier	12,622	215.6M	5.6 (5.0, 6.1)
Any private	Dental barrier	12,625	215.6M	9.2 (8.4, 9.9)
Any private	Prescription barrier	12,624	215.6M	2.5 (2.2, 2.8)
Any private	Zero OOP	12,723	217.3M	19.8 (18.7, 21.0)
Any private	Any barrier among zero-OOP	2,233	42.6M	6.9 (5.6, 8.2)
Public only	Any barrier	7,422	92.3M	15.2 (13.9, 16.5)
Public only	Medical barrier	7,420	92.3M	4.4 (3.8, 5.1)
Public only	Dental barrier	7,414	92.2M	12.7 (11.5, 13.9)
Public only	Prescription barrier	7,412	92.2M	3.7 (3.1, 4.3)
Public only	Zero OOP	7,554	94.3M	43.3 (41.3, 45.4)
Public only	Any barrier among zero-OOP	2,876	39.8M	7.1 (5.7, 8.6)
Uninsured	Any barrier	1,447	21.0M	30.0 (26.0, 34.0)
Uninsured	Medical barrier	1,444	20.9M	18.5 (15.8, 21.3)
Uninsured	Dental barrier	1,443	20.9M	23.9 (20.2, 27.7)
Uninsured	Prescription barrier	1,444	20.9M	6.4 (4.8, 8.1)
Uninsured	Zero OOP	1,470	21.5M	57.6 (53.8, 61.4)
Uninsured	Any barrier among zero-OOP	834	12.0M	22.5 (17.4, 27.5)

Table V6. Controlled descriptive-model robustness table for access barriers. Entries are weighted linear probability model coefficients in percentage points with HC1 robust standard errors in parentheses. The models are descriptive, not causal. The table tests whether the main gradient is only a one-way tabulation or persists after adding poverty, insurance, health, and demographic controls.

Variable / statistic	M1 burden quartiles only	M2 + poverty/insurance/health	M3 full controls	M4 threshold profile
Burden Q4	12.99 (0.95)	14.45 (0.94)	12.44 (1.07)	—
OOP ≥10%	—	—	—	11.48 (2.21)
Zero OOP	—	—	—	-9.52 (0.61)
Uninsured	—	17.08 (1.71)	17.72 (1.70)	17.81 (1.60)
Fair/poor health	—	12.50 (1.29)	12.21 (1.29)	14.45 (1.21)
High income	—	-11.99 (1.45)	-11.61 (1.45)	-9.76 (1.21)
Public only	—	0.34 (0.86)	1.96 (0.90)	-0.31 (0.73)
Unweighted N	17,043	17,043	17,043	20,595
Weighted population	256.6M	256.6M	256.6M	318.8M
Mean dependent variable	16.78	16.78	16.78	14.35
Weighted R-squared	0.022	0.066	0.084	0.062

The denominator table shows that very-low-income denominators inflate some poor/negative-income tail ratios, but the broad pattern is not erased by conservative floors: lower-resource groups still show higher burden and access-barrier rates. The component-CI table strengthens the zero-spending warning: among uninsured people with zero observed OOP spending, 22.5 percent report an access barrier, with an approximate 95 percent interval of 17.4 to 27.5 percent. The model robustness table reinforces the descriptive interpretation. Highest-burden-quartile, uninsured status, and fair/poor health remain strongly associated with access barriers after adding poverty, insurance, health, demographic, employment, family-size, race/ethnicity, and region controls. These estimates are still associations, not treatment effects.

4.11 Multi-year robustness feasibility

A further pass checks adjacent MEPS Full-Year Consolidated public-use files for 2019 through 2023. The purpose is limited: test whether the paper's 2022 measurement claim is a one-year artifact before treating multi-year evidence as submission-quality trend evidence.

Table M1. MEPS 2019–2023 variable continuity manifest. Full-Year Consolidated public-use files are audited for the core medical-fragility constructs used in the 2022 working paper. Variable names are listed when available; missing variables are shown as dashes.

Construct	2019	2020	2021	2022	2023
afrdca	AFRDCA42	AFRDCA42	AFRDCA42	AFRDCA42	AFRDCA42
afrddn	AFRDDN42	AFRDDN42	AFRDDN42	AFRDDN42	AFRDDN42
afrdpm	AFRDPM42	AFRDPM42	AFRDPM42	AFRDPM42	AFRDPM42
dlayca	DLAYCA42	DLAYCA42	DLAYCA42	DLAYCA42	DLAYCA42
dlaydn	DLAYDN42	DLAYDN42	DLAYDN42	DLAYDN42	DLAYDN42
dlaypm	DLAYPM42	DLAYPM42	DLAYPM42	DLAYPM42	DLAYPM42
faminc	FAMINC19	FAMINC20	FAMINC21	FAMINC22	FAMINC23
health	RTHLTH53	RTHLTH53	RTHLTH53	RTHLTH53	RTHLTH53
insurance	INSCOV19	INSCOV20	INSCOV21	INSCOV22	INSCOV23
oop	TOTSLF19	TOTSLF20	TOTSLF21	TOTSLF22	TOTSLF23
poverty	POVCAT19	POVCAT20	POVCAT21	POVCAT22	POVCAT23
varpsu	VARPSU	VARPSU	VARPSU	VARPSU	VARPSU
varstr	VARSTR	VARSTR	VARSTR	VARSTR	VARSTR
weight	PERWT19F	PERWT20F	PERWT21F	PERWT22F	PERWT23F

Table M2. MEPS 2019–2023 medical-fragility robustness benchmarks. Person-weighted estimates from Full-Year Consolidated public-use files. Bracketed entries use an approximate Taylor-style `VARSTR`/`VARPSU` linearization. The joint profile flags OOP ≥10 percent of family income, any delayed/unaffordable medical/dental/prescription care, or uninsured status.

Year	N	Weighted pop.	Any access barrier %	OOP ≥10% %	Zero OOP + barrier %	Joint profile %	Uninsured %	Uninsured barrier % [CI]	Barrier among zero-OOP % [CI]	P90 burden %
2019	28,512	323.1M	18.7	3.1	3.6	24.8	6.3	40.5 [37.1, 43.9]	13.2 [12.1, 14.3]	3.37
2020	27,805	324.8M	15.1	3.0	2.9	21.7	6.5	31.5 [28.1, 34.8]	10.0 [9.0, 11.1]	3.17
2021	28,336	327.2M	13.0	3.2	2.2	19.9	6.1	26.5 [23.0, 30.0]	7.8 [7.0, 8.6]	3.46
2022	22,431	328.9M	14.5	3.1	2.6	21.4	6.4	30.0 [26.0, 34.0]	9.0 [7.7, 10.2]	3.30
2023	18,919	330.0M	15.6	2.8	3.1	22.4	6.3	28.7 [25.0, 32.5]	11.3 [10.1, 12.6]	3.22

The continuity audit is encouraging. The core public-use constructs are available across 2019–2023: out-of-pocket medical spending, family income, poverty category, insurance status, latest-round health status, delayed/unaffordable medical, dental, and prescription care, person weights, and design variables. The benchmark table also supports the main measurement claim. Across all five years, uninsured people have much higher access-barrier rates than the population average, and a nontrivial zero-OOP group reports delayed or unaffordable care. That pattern is exactly why realized spending alone is not a sufficient fragility measure.

The table should not be overread as a finished trend analysis. Access-barrier rates are higher in 2019 than in 2021–2023, and the 2020–2021 pandemic period may affect utilization, access, and survey response. A journal version should add an item-by-item documentation appendix and check questionnaire/timing continuity before making strong time-series claims. For the current working paper, the result is enough to say that the 2022 finding is not isolated: the spending/access ambiguity appears across adjacent public-use MEPS years.

4.12 Additional uncertainty and robustness checks

A final May 17 review pass adds five additional checks. The purpose is practical: identify whether the measurement paper is strong enough to serve as a serious public working paper, and where it still needs specialized MEPS work before a submission-grade version. The answer is mixed in a useful way. The main person-weighted spending/access result is now well supported; the family and premium-inclusive extensions remain deliberately treated cautiously.

Table H1. MEPS 2022 survey-design uncertainty check. Estimates use person weights and the public `VARSTR`/`VARPSU` design variables. The table reports design SEs, a weighted-naive comparison, and the number of strata/PSUs actually contributing to the Taylor-style linearization. This is a transparency check, not a substitute for official AHRQ replicate guidance where required.

Group	Measure	N	Weighted pop.	Estimate %	Design SE pp	Approx. 95% CI	Design effect	Used/total strata	PSUs
All persons	Any access barrier	21,497	328.9M	14.5	0.42	13.6, 15.3	1.90	105/105	364
All persons	OOP ≥10% income	20,937	324.4M	3.1	0.16	2.8, 3.4	1.07	105/105	364
All persons	Zero OOP + barrier	21,497	328.9M	2.6	0.19	2.2, 2.9	1.86	105/105	364
Any private	Any access barrier	12,628	215.6M	12.6	0.45	11.7, 13.5	1.52	105/105	360
Any private	OOP ≥10% income	12,573	215.4M	2.4	0.16	2.1, 2.7	0.84	105/105	359
Any private	Zero OOP + barrier	12,628	215.6M	1.4	0.13	1.1, 1.6	1.04	105/105	360
Public only	Any access barrier	7,422	92.3M	15.2	0.68	13.9, 16.5	1.58	105/105	356
Public only	OOP ≥10% income	6,980	88.4M	4.7	0.34	4.1, 5.4	1.05	105/105	355
Public only	Zero OOP + barrier	7,422	92.3M	3.1	0.32	2.5, 3.7	1.52	105/105	356
Uninsured	Any access barrier	1,447	21.0M	30.0	2.03	26.0, 34.0	1.79	94/104	267
Uninsured	OOP ≥10% income	1,384	20.6M	3.5	0.79	1.9, 5.0	1.61	93/104	266
Uninsured	Zero OOP + barrier	1,447	21.0M	12.8	1.48	9.9, 15.7	1.79	94/104	267
Poor/negative income	Any access barrier	3,571	37.6M	20.4	1.09	18.3, 22.6	1.45	104/105	321
Poor/negative income	OOP ≥10% income	2,816	29.7M	10.9	0.82	9.2, 12.5	1.12	104/105	304
Poor/negative income	Zero OOP + barrier	3,571	37.6M	4.9	0.57	3.8, 6.1	1.40	104/105	321
High income	Any access barrier	7,998	141.5M	10.1	0.50	9.1, 11.0	1.45	105/105	348
High income	OOP ≥10% income	8,050	142.3M	0.9	0.12	0.7, 1.2	0.82	105/105	348
High income	Zero OOP + barrier	7,998	141.5M	1.0	0.14	0.7, 1.3	1.06	105/105	348

Table H2. MEPS 2022 family construction check. The provisional family key produces 11,244 collapsed family records from 22,431 persons. Family-size mismatches affect 1,752 person records; 1,211 collapsed records have missing/zero family weights. The table separates person-weighted and family-weighted estimands and keeps family results secondary until official family-estimation guidance is fully implemented.

Poverty/income group	Persons	Families	Positive-weight families	Missing/zero family weight	Weighted persons	Weighted families	Implied persons/family
Poor/negative	3,626	2,100	1,762	338	38.3M	18.7M	2.06
Near poor	1,017	568	508	60	12.5M	5.5M	2.26
Low income	3,000	1,579	1,422	157	41.2M	18.1M	2.27
Middle income	6,054	3,012	2,718	294	98.7M	41.7M	2.37
High income	8,050	3,985	3,623	362	142.3M	58.5M	2.44

Table H3. PRPL premium annualization checks. The processed premium field is treated as annual out-of-pocket premium exposure (`OOPX12X`). The final column is a deliberate failure-mode check: if an already annual premium were mistakenly multiplied by 12, implausibly high burden rates would result. This supports keeping PRPL premium-inclusive estimates exploratory and treated cautiously.

Poverty/income group	N	Weighted pop.	Median premium/income %	P90 premium/income %	Premium ≥10% %	Premium ≥20% %	Median OOP+premium/income %	P90 OOP+premium/income %	If x12 again: ≥50% %
Poor/negative	389	5.2M	13.07	108.00	54.3	41.3	18.40	144.83	63.5
Near poor	166	2.4M	4.51	24.71	33.9	17.7	6.88	44.02	54.5
Low income	852	12.9M	5.93	23.36	33.2	13.5	8.22	28.85	60.3
Middle income	3,281	56.1M	4.24	11.55	14.2	2.8	5.23	14.71	50.7
High income	6,120	111.2M	1.98	6.26	3.2	0.5	2.63	8.02	21.2

Table H4. MEPS 2019–2023 questionnaire/variable continuity audit. Core public-use constructs used in the working paper are available in all five adjacent Full-Year Consolidated files. Exact names vary for annual measures by year suffix, while access items use stable round-specific names. This supports robustness benchmarking but still calls for a formal item-wording appendix before strong trend claims.

Construct	Years present	Exact name stable	2019 variable	2023 variable
afrdca	5/5	Yes	AFRDCA42	AFRDCA42
afrddn	5/5	Yes	AFRDDN42	AFRDDN42
afrdpm	5/5	Yes	AFRDPM42	AFRDPM42
dlayca	5/5	Yes	DLAYCA42	DLAYCA42
dlaydn	5/5	Yes	DLAYDN42	DLAYDN42
dlaypm	5/5	Yes	DLAYPM42	DLAYPM42
faminc	5/5	No	FAMINC19	FAMINC23
health	5/5	Yes	RTHLTH53	RTHLTH53
insurance	5/5	No	INSCOV19	INSCOV23
oop	5/5	No	TOTSLF19	TOTSLF23
poverty	5/5	No	POVCAT19	POVCAT23
varpsu	5/5	Yes	VARPSU	VARPSU
varstr	5/5	Yes	VARSTR	VARSTR
weight	5/5	No	PERWT19F	PERWT23F

Table H5. Benchmark positioning against underinsurance-style screens. The joint profile is intentionally broader than a spending-only burden threshold. The private underinsurance-style screen approximates privately insured people who either meet the OOP ≥10 percent threshold or report an access barrier. This table positions the proposed profile against familiar burden, access, and coverage screens without claiming a final underinsurance definition.

Screen	Weighted share %	Classified pop.	Access among classified %	Zero OOP among classified %	Uninsured among classified %
High OOP burden only	3.1	10.1M	33.6	0.0	7.1
Any access barrier	14.5	47.6M	100.0	17.8	13.2
Uninsured	6.4	21.5M	30.0	57.6	100.0
Private underinsurance-style screen	9.7	31.0M	88.1	9.5	0.0
Access barrier missed by high-burden screen	13.3	42.7M	100.0	18.8	13.2
Zero OOP with access barrier	2.6	8.5M	100.0	100.0	31.8
Joint fragility profile	21.4	69.2M	69.4	26.2	31.0

These checks clarify how much weight the paper can carry. The selected survey-design estimates now show strata and PSU coverage rather than only point estimates. The family construction table explains why family-weighted results should remain secondary until rebuilt under official MEPS guidance. The PRPL premium check shows that premium-inclusive burden is promising but fragile, especially for lower-income privately insured families and for any annualization mistake. The 2019–2023 continuity table supports adjacent-year robustness while still requiring an item-wording appendix before strong trend claims. Finally, the benchmark table shows the substantive payoff: a spending-only threshold identifies a much smaller population than a profile that also captures access barriers and uninsured status.

5. Conditional descriptive model

Table 9 reports the main weighted linear probability model for any delayed or unaffordable medical, dental, or prescription care; Table V6 above reports a compact model robustness table with HC1 robust standard errors. The model includes burden quartile, poverty category, insurance status, fair/poor health, chronic-condition count, age, sex, employment, family size, race/ethnicity codes, and region. It is not a causal model. It describes how dimensions of risk cluster conditional on observed covariates.

Table 9. Weighted linear probability model for any medical/dental/prescription access barrier. Dependent variable is any delayed or unaffordable care indicator. Coefficients are percentage points. Reference categories: OOP burden Q1, poor/negative income, any private insurance, excellent/very good/good health, Hispanic, and Northeast. This is a descriptive clustering model, not a causal effect estimate; a journal version should add MEPS survey-design standard errors.

Variable	Coefficient pp / statistic
Intercept	20.74
Burden Q2	3.87
Burden Q3	7.10
Burden Q4	12.44
Near poor	-0.48
Low income	-2.21
Middle income	-5.50
High income	-11.61
Public only	1.96
Uninsured	17.72
Fair/poor health	12.21
Chronic condition count	1.92
Age/10	-1.47
Female	3.32
Employed	4.26
Family size	-2.52
Non-Hispanic White only	-0.25
Non-Hispanic Black only	-1.62
Non-Hispanic Asian only	-3.26
Non-Hispanic other/multiple race	-3.70
Midwest	2.09
South	2.54
West	5.09
Mean dependent variable	16.78

Variable	Coefficient pp / statistic
Unweighted N	17,043
Weighted population	256.6M
Weighted R-squared	0.084

The key associations survive richer controls. The highest burden quartile is associated with a 12.44 percentage point higher access-barrier probability relative to the lowest quartile. Uninsured status is associated with 17.72 percentage points higher probability relative to private insurance. Fair or poor health is associated with 12.21 percentage points higher probability. These estimates should be interpreted as conditional co-occurrence, not as effects.

5.1 Benchmarking the joint fragility profile against simpler screens

A measurement paper needs to show that the proposed profile adds something beyond familiar measures. Table 10 therefore compares the joint medical-financial fragility profile with simpler screens: OOP at least 10 percent of income, any access barrier, uninsured status, burden or access, access barriers missed by the high-burden screen, and zero OOP with an access barrier.

The joint profile is intentionally broad. It flags people with OOP ≥10 percent of income, any delayed or unaffordable care barrier, or uninsured status. This is not a causal risk score and not a final underinsurance definition. It is a descriptive screen designed to prevent a spending-only measure from treating foregone care as protection.

Table 10. Benchmarking the joint medical-financial fragility profile against simpler screens. The proposed joint profile is intentionally broader than a realized-spending threshold: it flags people with OOP ≥10 percent of income, any delayed/unaffordable care barrier, or uninsured status. The comparison shows how many people are missed by spending-only measures and why zero OOP cannot be treated as protection.

Measure	Weighted share %	Classified pop.	Diagnostic composition among classified	Interpretation
OOP ≥10% of income	3.1	10.1M	Access 33.6%; OOP≥10 100.0%; Zero OOP 0.0%; Fair/poor health 25.5%	Standard high-burden screen among positive-income records.
Any access barrier	14.5	47.6M	Access 100.0%; OOP≥10 7.3%; Zero OOP 17.8%; Fair/poor health 21.4%	Delayed/unaffordable medical, dental, or prescription care.
Uninsured	6.4	21.5M	Access 30.0%; OOP≥10 3.5%; Zero OOP 57.6%; Fair/poor health 8.8%	Annual uninsured status only.
OOP ≥10% OR access barrier	16.9	54.4M	Access 87.8%; OOP≥10 19.2%; Zero OOP 15.6%; Fair/poor health 21.4%	Combines realized burden and unmet-need/access signal.
Joint fragility profile	21.4	69.2M	Access 69.4%; OOP≥10 15.1%; Zero OOP 26.2%; Fair/poor health 18.3%	OOP ≥10%, any access barrier, or uninsured.
Access barrier missed by high-burden screen	13.3	42.7M	Access 100.0%; OOP≥10 0.0%; Zero OOP 18.8%; Fair/poor health 19.8%	Access barrier with OOP below 10% of income.
Zero OOP with access barrier	2.6	8.5M	Access 100.0%; OOP≥10 0.0%; Zero OOP 100.0%; Fair/poor health 12.8%	Foregone/delayed-care risk invisible in spending alone.

The benchmark comparison is the clearest answer to the paper's contribution question. A high-burden screen identifies about 10.1 million people, while an access-barrier screen identifies 47.6 million people and the joint profile identifies 69.2 million people. About 42.7 million people report an access barrier while not meeting the OOP ≥10 percent threshold, and about 8.5 million report both zero OOP and an access barrier. Those are the households a realized-spending threshold is most likely to misread.

6. Measurement lessons

The first lesson is that realized spending is ambiguous. The uninsured have lower observed out-of-pocket spending than privately insured people but much higher access barriers. Zero spending is not necessarily protection.

The second lesson is that burden shares and threshold measures tell a different story than dollar spending. Higher-income people spend more dollars, but lower-income people face higher resource pressure.

The third lesson is that access outcomes should be co-primary in medical financial fragility measurement. A spending-only analysis will miss unmet need. A coverage-only analysis will miss plan generosity and cost-sharing exposure. An access-only analysis will miss the financial channel. MEPS can observe all three, though not perfectly.

The fourth lesson is that family-level construction and premium construction are not optional details. They determine the estimand. The new audit tables make the current draft more transparent, but a credible journal version still needs MEPS-guidance validation for family burden and PRPL premiums.

The fifth lesson is that the joint profile adds information beyond a high-spending screen. The spending-only OOP ≥10 percent threshold identifies about 10.1 million people, while the joint profile identifies 69.2 million. Most importantly, 42.7 million people report an access barrier while not meeting the high-burden threshold, and 8.5 million people report both zero OOP and an access barrier. Those are precisely the cases that motivate measuring burden, coverage, need, and access together.

7. Limitations and next steps

The paper has five important limitations.

First, it is descriptive. Medical spending, health, insurance, income, and access barriers are jointly determined.

Second, the current build reports approximate MEPS design-based confidence intervals and now documents strata/PSU reliability for selected descriptive proportions. The next revision must extend design-based inference to means, threshold estimates, and model coefficients, and verify the implementation against AHRQ/MEPS variance guidance or official replicate guidance where applicable.

Third, the income denominator remains imperfect. Future versions should add poverty-threshold-normalized denominators, winsorized ratios, zero/negative-income sensitivity, and more explicit treatment of disposable resources.

Fourth, premium-inclusive estimates remain exploratory. The PRPL algorithm now has linkage sensitivity and annualization guardrail checks, but it still needs validation against official MEPS PRPL documentation before premium-inclusive results can carry headline weight.

Fifth, the multi-year evidence is a robustness benchmark rather than a full trend design. The 2019–2023 continuity audit is encouraging, but a final version should add an item-wording and timing appendix before making strong time-series claims.

8. Conclusion

This paper reframes medical affordability as a household financial-fragility measurement problem. Its core claim is not that lower-income, uninsured, or sicker people face more difficulty accessing care; that is already known. Its narrower contribution is to show why cost burden, insurance design, health need, and access barriers should be measured jointly.

The evidence supports that measurement frame. Lower-income groups have higher burden shares and more access barriers. Poor-health groups face especially high burden and barriers. Uninsured people have high access barriers despite lower observed out-of-pocket spending. Zero observed spending can coexist with unmet need. The benchmark comparison makes the contribution sharper: a high-burden spending threshold misses many people with delayed or unaffordable care, while the joint profile captures the access and coverage margins that spending alone cannot see.

The current draft is therefore a stronger working-paper version of a promising MEPS measurement project. The next milestone is empirical validation: broader design-based inference, MEPS-guidance validation of family construction, PRPL annualization checks, questionnaire-continuity review, and deeper benchmarking against standard underinsurance and cost-related-access measures.

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